Name+Email ID

CSCI 5521: Machine Learning Fundamentals (Spring 2022)

Quiz 3 (Thurs, Mar 31))

Due on Gradescope at 02:00 PM, Friday, Apr 1

Instructions:

- This quiz has 3 questions, 30 points, on 1 page.
- Please **write your name & ID on this cover page**.

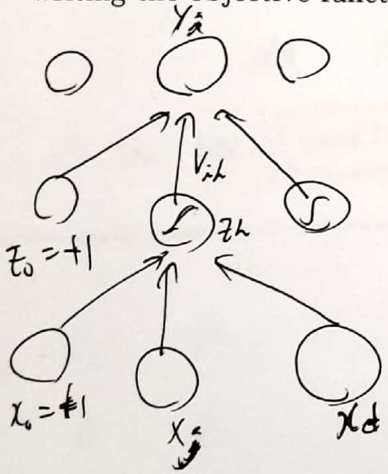
1. (**9 points**) Select all the correct statement(s). (a), (b), (c)

   (a) A Perceptron (without activation functions) is a linear model.

   (b) A Multilayer Perceptron (without activation functions) is a linear model.

   (c) A Perceptron can be used for both classification and regression.

2. (**13 points**) Pick one parameter in a (Multilayer) Perceptron, and write its update equation. The parameter can be for a node at any layer and for either classification or regression. Start with writing the objective function.



output layer $y_i = v_i^T z = \sum_{h=1}^{H} v_{ih} z_h + v_{i0}$

$E(W, v \mid X) = \frac{1}{2} \sum_t (r^t - y^t)^2$

hidden layer $\therefore \Delta v_h = -\eta \frac{\partial E}{\partial v_h} = \eta \sum_t (r^t - y^t) z_h^t$

3. (**8 points**) Give an example of an activation function in Multilayer Perceptron. You can write either the name or the mathematical representation of the function. Briefly discuss roles of activation functions in a Multilayer Perceptron.

sigmoid.

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

weight sum gives any number, but activation function decides a neuron (or perceptron) to be activated or not. It decides whether neuron's input to the neuron is important or not. So, sigmoid func put the weighted sum into the range (squishes) between 0 & 1. { = negative inputs ⇒ close to 0; positive inputs ⇒ close to 1

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